

Appendix E: Solutions' Reference Designs, & Research

[1] Reference design inspiring the one-way flap: One way pet doors [11, Bibliography]

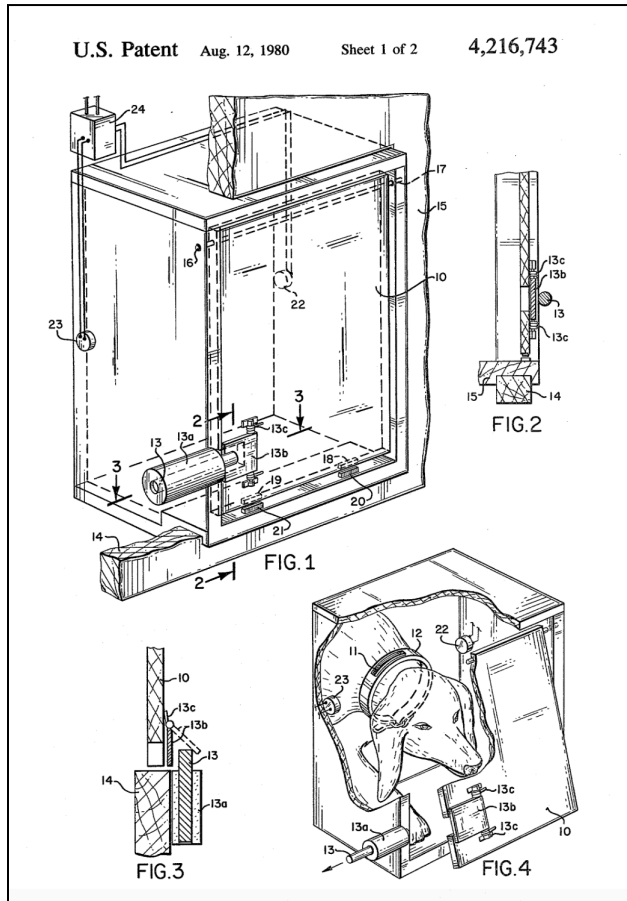
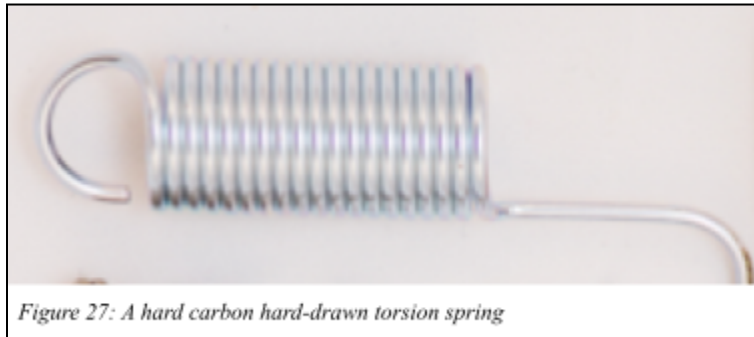


Figure 21: Patent drawing of a one-way pet door, hinged at the top to open in both directions for ingress and egress, but is prevented from opening in the ingress direction by a one-way latch

This reference design helped to inspire the one-way flap design of the Snack Chute. The Snack Chute utilizes a similar flap which is hinged and is able to open in both directions, however is prevented from opening in one direction to achieve the anti-pilfer goal.

[2] Research for the spring choice of the Snack Chute in the Three Key Decisions [13, Bibliography]



Torsion springs are used for bending purposes with common materials including:

- Alloy Steel: for high strength and durability
- Plastics: for extremely light-duty needs
- Hard-drawn steel: for general-use where high strength and durability are not required

Since the spring must handle repetitive movement, so plastic cannot be chosen, and the mass of the one-way-flap and snacks, which are lightweight and alloy steel cannot be chosen, therefore hard-drawn steel is chosen.

[3.1] Reference design inspiring the pouch: Skateboard ramp [14, Bibliography]

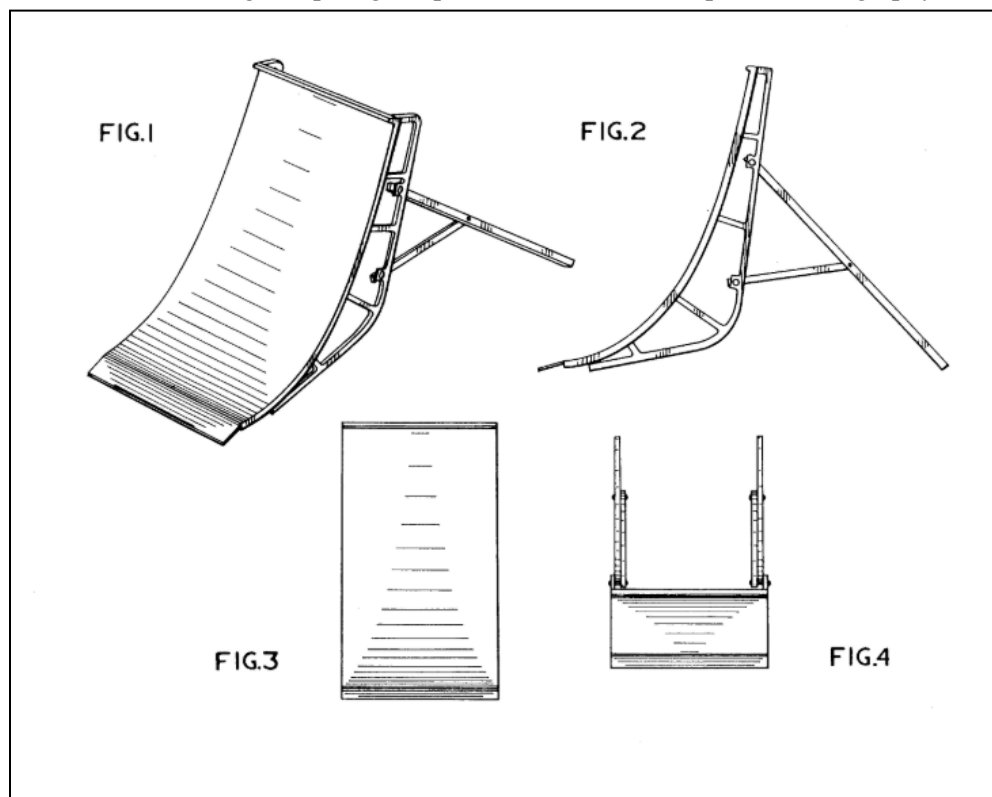


Figure 22: Patent drawing of a compact skateboard ramp, whose smooth curved profile inspired the inclined slope in Snack Sack, guiding snacks along a continuous surface toward the user.

The skateboard ramp patent provides a compact, smoothly curved ramp geometry that allows a skateboard to roll from a higher platform down to ground level without sharp corners or sudden drops. We used this as a reference design for the Snack Sack slope: the same gradual curved transition is applied inside the vending bin so that snacks slide along a continuous surface, rather than bouncing or sticking on a flat ledge, and are guided to a predictable, higher resting point near the user within the 170 mm reach range.

[3.2] Reference design mentioned in Snack Sack: Ultrahigh molecular weight polyethylene (UHMWPE) [15, Bibliography]



Figure 23: Example of a UHMWPE bushing, used in Snack Sack at the flap's pivot to provide a low-friction, wear-resistant bearing surface that increases the flap's opening angle.

We chose ultrahigh molecular weight polyethylene (UHMW-PE) for the bushings because it is explicitly designed for low-friction sliding contacts such as bearings and bushings: it has a very low coefficient of friction, excellent abrasion resistance, and is widely used in conveyor and material-handling bushings for exactly this reason. Together, these properties make UHMW-PE a good candidate for reducing hinge friction while keeping the retrofit simple and durable.

[4] Reference design inspiring the sensor: Sensor operated trash can [16, Bibliography]

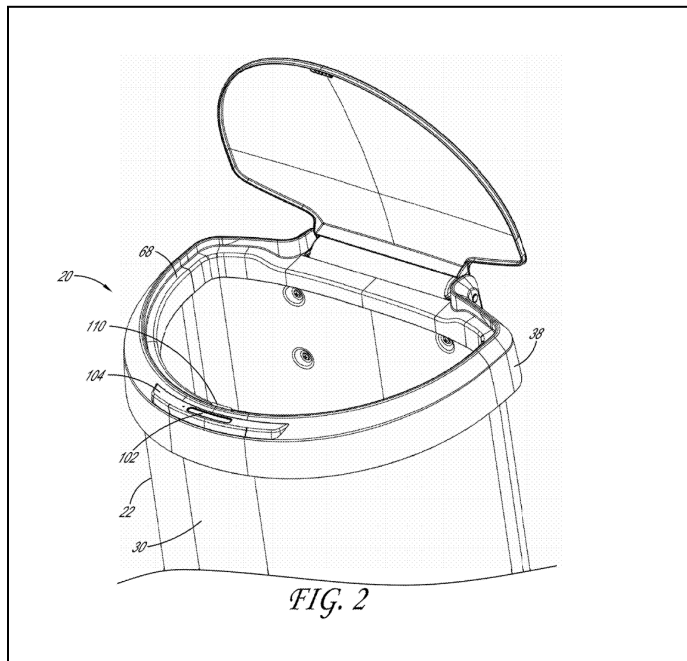


Figure 24: Patent drawing of a sensor-operated trash can illustrating its automated lid operation mechanism.

This diagram shows the functionality of a sensor-activated trash can, including its automatic lid opening mechanism and visual indicator system. The trash can's use of a blinking green light to signal sensor activation and a blinking red light to warn for the closure of the lid directly influenced the design of our sensor operated vending machine flap door. While both systems (garbage can and vending machine) rely on visual cues and automated movement to improve the ease of user interaction, the two systems are not exactly the same. The garbage can is automated such that the sensor is placed directly underneath the user's field of view and the flap opens outwards rather than inwards like the vending machine does.

[5] Reference design inspiring the Lift n' Pick: Garage door [17, Bibliography]

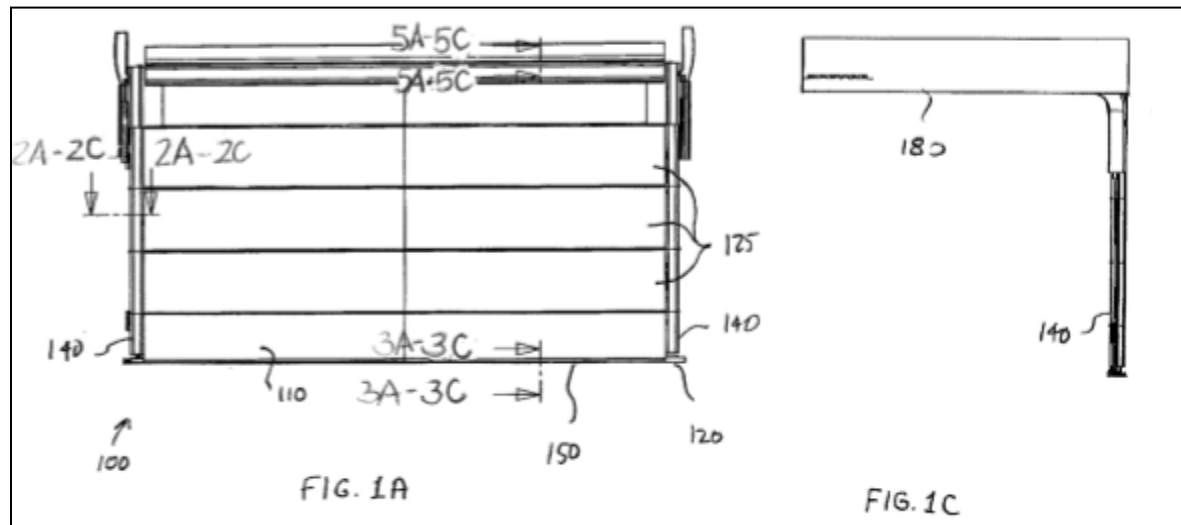


Figure 25. A garage door sealing system illustrating a closed garage door on the left and the path the garage door would follow when opened.

This diagram displays the opening and sealing mechanism of a garage door from the front and side view. The moving mechanism for opening a garage door is similar to the mechanism inspired by the vertical sliding door for Design 4. As the garage opens, the metal from the door is shifted to the top, blocking the horizontal ceiling, which in the case of the vending machine, blocks access to the stored snacks above. Although the purpose of the garage door and vending machine sliding door are both for opening and enclosing a space, the garage door is significantly heavier and larger, so it is difficult to be opened by human wrists. Therefore, the garage door fails to address the force requirement of the vending machine opening system.

[6] Reference design for proxy testing of Lift n' Pick: Suitcase handle [23, Bibliography]

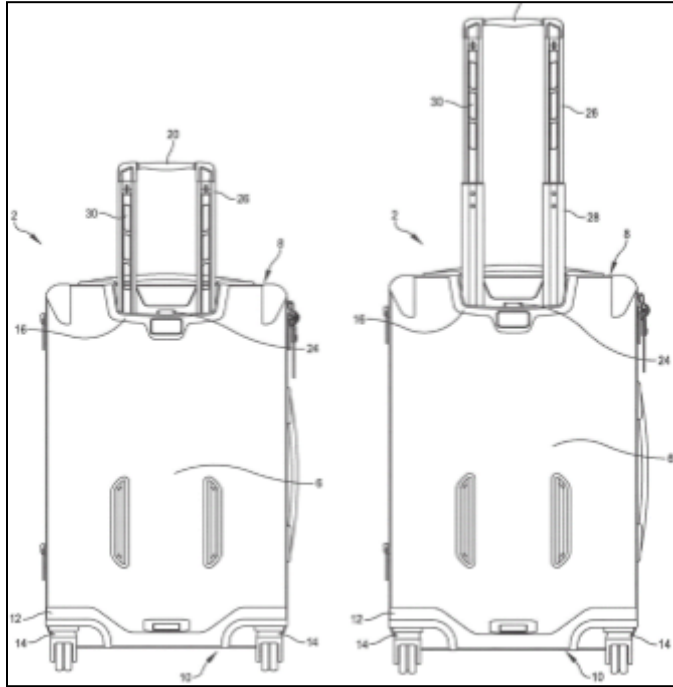


Figure 26. A patent of extendable handles of a suitcase showing varying heights of the handle.

This diagram shows the extendable handle features of the suitcase. The vertical wrist movement necessary for extending this suitcase handle is similar to the motion of vertically pulling on a sliding door. Therefore, a suitcase is used to model and measure this motion during our testing. However, the suitcase handle is designed for elevating height to make luggage easier to pull, while the sliding door is designed for opening and enclosing a space. Therefore, the suitcase handle fails to address the open and closing purpose of the vending machine slider.